JABstem
Not to be shared without copyright holder's permission

Past Papers
Nat 5
Physics
2024
Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	/125	%	
A	86+	69%	34.4%
B	73+	58%	21.4%
C	60+	48%	18.0%
D	47+	38%	13.8%
No award	<47	<38%	12.3%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.0 /25	41.2 /75	17.386 /25

2024 Nat5 Physics Marking Scheme

Question	Answer	% Correct	Physics Covered

Question 1 — Answer E — 64%

Vector Quantity	force	velocity	displacement	acceleration	**weight**
Scalar Quantity	**energy**	**speed**	distance	time	mass

Question 2 — Answer B — 75%

Average speed between X and Y	**Instantaneous speed at Y**
average speed = $\frac{\text{total distance}}{\text{time taken}}$	instantaneous speed = $\frac{\text{length of card}}{\text{time to pass light gate}}$
average speed = $\frac{d}{t_1}$	instantaneous speed = $\frac{l}{t_2}$

Question 3 — Answer E — 58%

☒A The force of the Earth on exhaust gases is not the reaction force in example.
☒B The force of the Earth on the rocket engines is not the reaction force in example.
☒C The force of the rocket engines on Earth is not the reaction force in example.
☒D The force of the exhaust gases on Earth is not the reaction force in example.
☑E The force of the rocket engines on the exhaust gases is equal and opposite to the force of the exhaust gases on the rocket engines (Newton's Third Law)

Question 4 — Answer B — 64%

E_p = ? m = 2400kg g = 9.8 N kg^{-1} h = 5.0m

E_p = m x g x h

E_p = 2400 x 9.8 x 5.0

E_p = 117600 J

E = 117600 J P = ? t = 2 minutes = 2x60s

$$P = \frac{E}{t} = \frac{117600}{2x60} = 980 \text{ W}$$

Question 5 — Answer A — 63%

Statement I - Correct	Statement II - Incorrect	Statement III - Incorrect
The horizontal speed of the satellite is constant but not the velocity as it is not travelling in a straight line	The satellite is moving in both the horizontal and vertical directions to stay in orbit	The is still gravitational field strength while in orbit around Earth.

Question 6 — Answer D — 39%

Time for laser light to travel to moon and back = 2.6s
Time for laser light to travel to moon and back = 1.3s

d = v x t

d = 3x10^8 m s^{-1} x 1.3s

d = 3.9x10^8 m

Question 7 — Answer C — 91%

AprizeSat-3 has a higher altitude (686km) than Hubble Space Telescope (537km)
∴ AprizeSat-3 must have period greater than 95 minutes
AprizeSat-3 has a lower altitude (696km) than Intelsat-18 (35800km)
∴ AprizeSat-3 must have a period less than 1440 minutes

Question 8 — Answer D — 81%

Statement I - Incorrect	Statement II - Correct	Statement III - Correct
A light year is the distance travelled by light in one year.	The approximate age of the universe is 13.8 billion years.	The Big Bang Theory describes the origin of the universe

Question 9 — Answer A — 47%

☑A Particle is attracted to Q & R ∴ Particle has opposite charge to particles Q & R
☒B Particle is attracted to Q ∴ Particle and Q cannot have same charge
☒C Particle is attracted to R ∴ Particle and R cannot have same charge
☒D Particle is attracted to R ∴ Particle and R cannot have same charge
☒E Particle is attracted to Q ∴ Particle and Q cannot have same charge

10 B 43

Statement I - Incorrect	Statement II - Correct	Statement III - Incorrect
1 Watt of power is the transfer of 1 Joule of energy per second $1.2kW = 1200W = 1200\ J\ s^{-1}$	Appliances over 720W require a 13A fuse to be fitted to the plug.	$P = 1.2kW = 1200W \quad I = ? \quad V = 230V$ $P = I \times V$ $1200 = I \times 230$ $I = {}^{1200}/_{230} = 5.22A$ $Q = ? \quad I = 5.22A \quad t = 1s$ $Q = I \times t$ $Q = 5.22 \times 1$ $Q = 5.22C$

11 E 63

$V_1 = 3.6V \qquad V_2 = ? \qquad R_1 = 16\Omega \qquad R_2 = 24\Omega$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2} \therefore \frac{3.6}{V_2} = \frac{16\Omega}{24\Omega} \therefore \frac{3.6 \times 24}{16} = V_2 \therefore V_2 = 5.4V$$

12 A 28

Reading on Ammeter A_1	Reading on Ammeter A_2	Reading on Ammeter A_3
Decreases	Decreases	Decreases
As light on LDR decreases the resistance of LDR increases. Total resistance in circuit increases and current in A_1 will decrease as result.	As current in A_1 decreases, the split of this lower current at the branches will give a lower current in A_2	As current in A_1 decreases, the split of this lower current at the branches will give a lower current in A_3

13 B 65

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{12} + \frac{1}{6.0} + \frac{1}{4.0} = \frac{1}{12} + \frac{2}{12} + \frac{3}{12} = \frac{6}{12} \therefore R_T = {}^{12}/_{6} = 2.0\Omega$$

14 C 86

$P = ? \qquad I = 1.5A \qquad R = 8.0\Omega$

$P = I^2 \times R$
$P = (1.5)^2 \times 8.0$
$P = (1.5)^2 \times 8.0$
$P = 2.25 \times 8.0$
$P = 18\ W$

15 B 84

☒A Heating block for longer increases temperature gained but not improve accuracy
☑B Insulation would reduce heat loss from block and improve accuracy
☒C A block of lesser mass would gain a higher temperature but not improve accuracy
☒D Higher power would increase temperature gained but not improve accuracy
☒E Removing heat proof mat leads to more energy being lost and decrease accuracy

16 D 62

☒A P→Q represents the increase in temperature in the solid state
☒B Q→R represents the latent heat of fusion from solid to liquid
☒C R→S represents the increase in temperature in the liquid state
☑D S→T represents the latent heat of vaporisation from liquid to gas
☒E T→U represents the increase in temperature in the gas state

17 D 52

$V_1 = 1.40\times10^{-6}\ m^3$	$p_1 = 120\ kPa$	Temperature constant
$V_2 = 1.40\times10^{-6}\ m^3 - 0.30\times10^{-6}\ m^3 = 1.1\times10^{-6}\ m^3$	$p_2 = ?$	

$p_1 \times V_1 = p_2 \times V_2$

$1.40\times10^{-6} \times 120 = p_2 \times 1.1\times10^{-6}$

$$\frac{1.40\times10^{-6} \times 120}{1.1\times10^{-6}} = p_2$$

$p_2 = 153\ kPa$

18 A 77

$F_b = 360$ N $p = 1020$ kg m^{-3} g = 9.8 N kg^{-1} V = ?

$$F_b = p \times g \times V$$

$$360 = 1020 \times 9.8 \times V$$

$$V = \frac{360}{1020 \times 9.8}$$

$$V = 0.036 \text{ m}^3$$

19 C 74

3 waves travel 24m at 28 m s^{-1} therefore $v = 48$ m s^{-1} d = 24 m t = ?

$$v = \frac{d}{t} \quad \therefore \quad 48 = \frac{24}{t} \quad \therefore \quad t = \frac{24}{48} = 0.5 \text{ s}$$

f = ? N = 3 t = 0.5 s

$$f = \frac{N}{t} = \frac{3}{0.5} = 6.0 \text{ Hz}$$

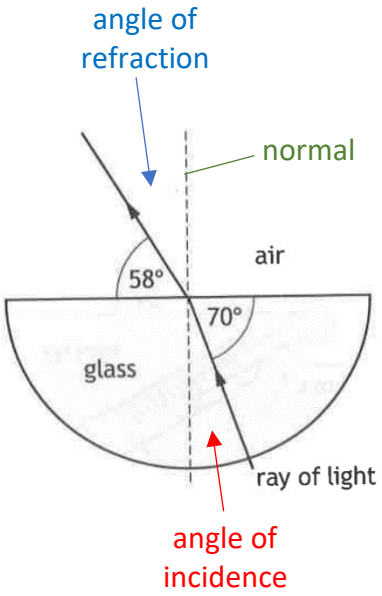
20 C 57

X	Y	Z
Longer	**More**	**The Same As**
Infrared has a longer wavelength than visible light	Longer infrared wavelengths diffract more than shorter visible wavelengths	Infrared and visible light are both EM radiation and both travel at 3x10^8 m s^{-1}

21 A 64

Angle of Incidence = 90° – 70° = 20°

Angle of Refraction = 90° – 58° = 32°



22 B 79

☒A Removing a proton would be a nuclear reaction and not part of ionisation
☑B Ionisation involves the removal of an electron to form a charged particle
☒C Removing a (neutral) neutron would not change the charge on the particle
☒D Nuclear Fission: Splitting a large nucleus into smaller nuclei
☒ E Nuclear Fusion: Joining of small nuclei to form a larger nucleus

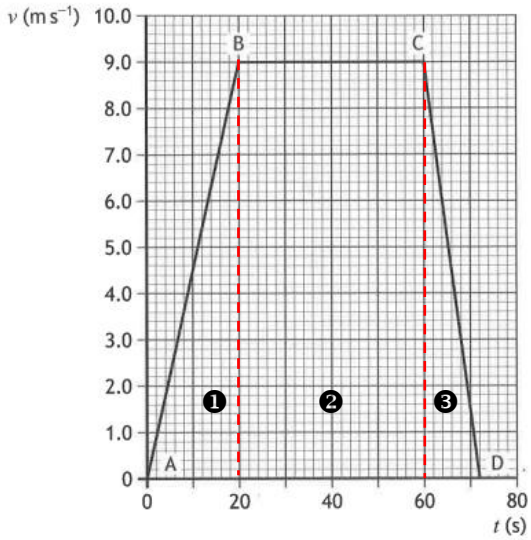
23 E 69

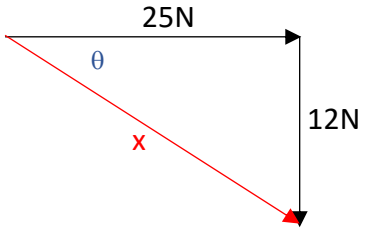
A = ? N = 1.8x10^6 nuclei decay t = 10 hours = 10 x 60 x 60 s

$$A = \frac{N}{t} = \frac{1.8 \times 10^6}{10 \times 60 \times 60} = 50 \text{ Bq}$$

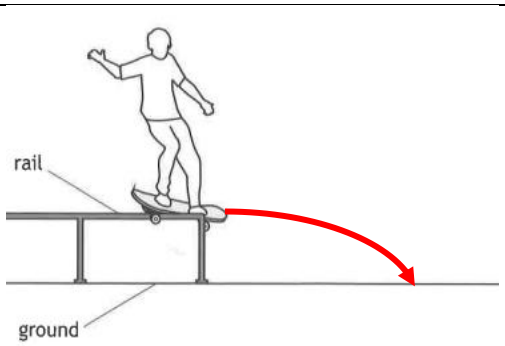
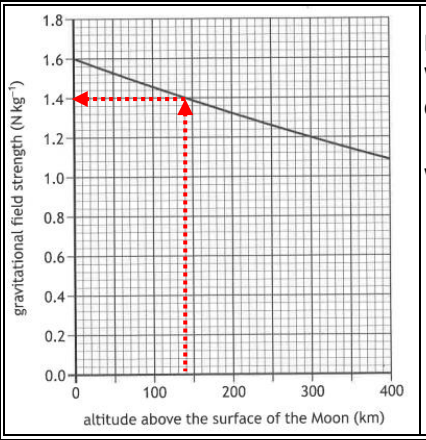
©JABstem 2024

24	C	63	H	=	$D w_r$ (alpha radiation)	+	$D w_r$ (slow neutrons)	
				=	(0.20 μGr x 20)	+	(6.0 μGr x 3)	
				=	4.0	+		
				=	18.6 μSv			
25	B	37	☒A Original reading of 400 counts s^{-1} has not halved to 200 counts s^{-1} by 20 minutes ☑B Increasing the time to 40 minutes would allow the count rate to fall below 200 counts per second allowing an easier measurement of half-life from the graph ☒C Lead shielding would stop/reduce the radiation reaching the Geiger-Müller tube ☒D Closer detector increases count rate but graph will not have halved in that time ☒E Further detector decreases count rate but graph will not have halved in that time					

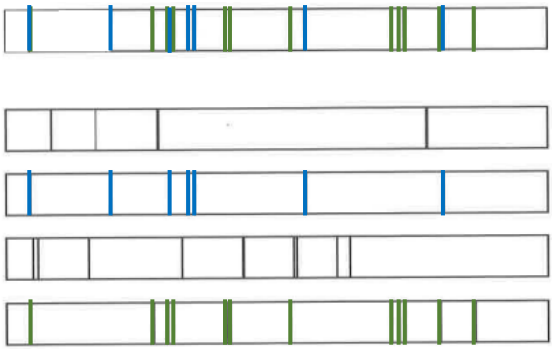
<table>
<tr><th>Question</th><th>Answer</th><th>Chemistry Covered</th></tr>
<tr><td>1a(i)</td><td>Constant Acceleration</td><td rowspan="2"><table>
<tr><td>When graph is uphill in a straight line</td><td>constant acceleration</td></tr>
<tr><td>When graph is horizontal in straight line</td><td>constant speed</td></tr>
<tr><td>When graph is downhill in a straight line</td><td>deceleration or negative acceleration</td></tr>
</table></td></tr>
<tr><td>1a(ii)</td><td>Constant speed</td></tr>
<tr><td>1b</td><td>- 0.75 m s^{-2}</td><td>a = ? v = 0.0 m s^{-1} u = 9.0 m s^{-1} t = 12 s
$a = \frac{v - u}{t} = \frac{0.0 - 9.0}{12} = -0.75$ m s^{-2}
(1 mark) (1 mark) (1 mark)</td></tr>
<tr><td>1c</td><td>504m</td><td> 

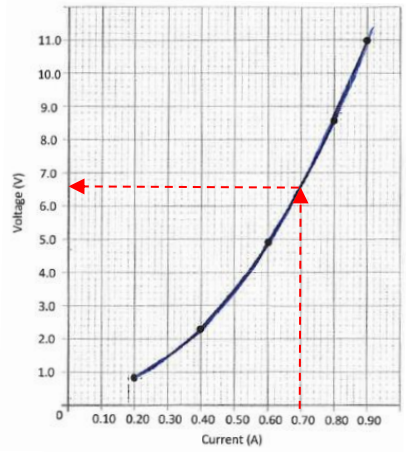
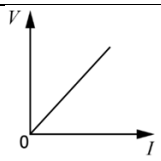
<table>
<tr><td>Area ❶</td><td>Area ❷</td><td>Area ❸</td></tr>
<tr><td>Distance = area under graph</td><td>Distance = area under graph</td><td>Distance = area under graph</td></tr>
<tr><td>= ½ x 20 x 9</td><td>= 40 x 9</td><td>= ½ x 12 x 9</td></tr>
<tr><td>= 90</td><td>= 360</td><td>= 54</td></tr>
<tr><td colspan="3">Total distance = 90 + 360 + 54 = 504m</td></tr>
</table></td></tr>
<tr><td>2a(i)A</td><td>28 N</td><td> 

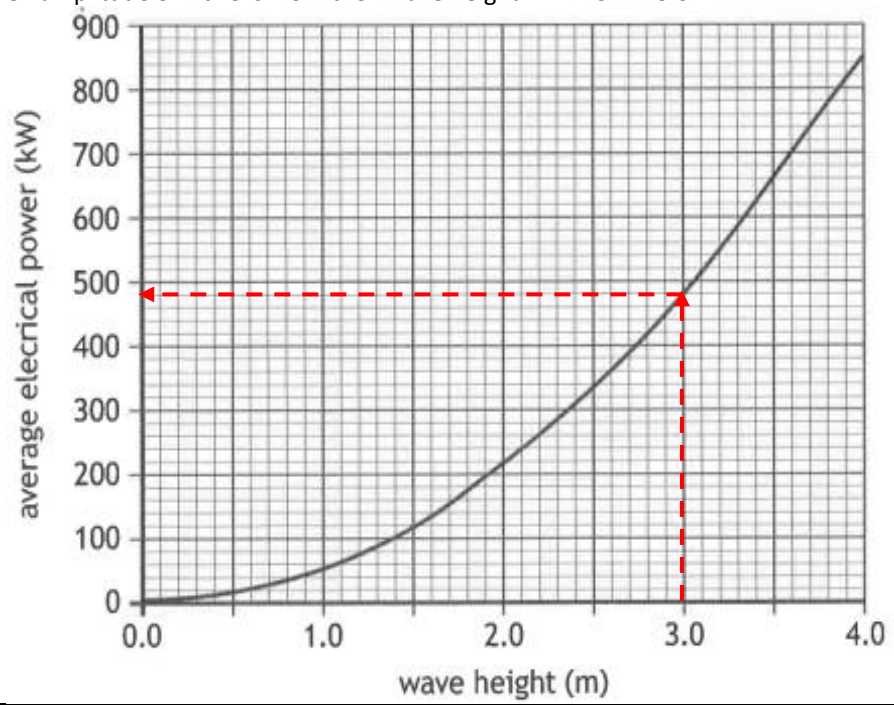
$x = \sqrt{(25)^2 + (12)^2}$
$x = \sqrt{625 + 144}$
$x = \sqrt{769}$
$x = 28$ N</td></tr>
<tr><td>2a(i)B</td><td>116</td><td>$\tan\theta = \frac{opp}{adj} = \frac{12}{25} = 0.48 \quad \therefore \theta = 26^o$
$\therefore$ bearing = 90° + 26° = 116</td></tr>
<tr><td>2a(ii)</td><td>0.37 m s^{-2} at 116</td><td>a = ? F = 28 N m = 75 kg
F = m a (1 mark)
28 = 75 x a (1 mark)
a = 0.37 m s^{-2} at 116 (1 mark)</td></tr>
<tr><td>2b</td><td>One answer from:</td><td><table>
<tr><td>adopt a streamlined position</td><td>cycle behind another triathlete</td><td>wear tight clothing/lycra</td><td>solid wheels instead of spokes</td></tr>
</table></td></tr>
<tr><td>2c(i)</td><td>4.4 m s^{-1}</td><td>d = 4 x 2.5 km = 4 x 2500 m = 10000 m $\bar{v}$ = ? t = 38 minutes = 38 x 60 s = 2280 s
d = $\bar{v}$ t (1 mark)
10000 = $\bar{v}$ x 2280 (1 mark)
$\bar{v} = {}^{10000}/_{2280} = 4.4$ m s^{-1} (1 mark)</td></tr>
</table>

<table>
<tr><td>2c(ii)</td><td>0 m s^{-1}</td><td>The total displacement after 4 laps back to the same point would be 0 m
$\therefore$ average velocity 0 m s^{-1}</td></tr>
<tr><td>3a(i)</td><td>1300J</td><td>At start:
E_k = ? m = 65 kg v = 7.0 m s^{-1}
$E_k = \frac{1}{2}$ m v^2
$E_k = \frac{1}{2}$ x 65 x $(7.0)^2$
E_k = 1592.5 J
At end:
E_k = ? m = 65 kg v = 3.0 m s^{-1}
$E_k = \frac{1}{2}$ m v^2
$E_k = \frac{1}{2}$ x 65 x $(3.0)^2$
E_k = 292.5 J
Decrease in Kinetic Energy = 1592.5 – 292.5 = 1300 J</td></tr>
<tr><td>3a(ii)</td><td>650 N</td><td>E_w = 1300 J F = ? d = 2.0m
E_w = F d (1 mark)
1300 = F x 2.0 (1 mark)
F = 650 N (1 mark)</td></tr>
<tr><td>3b</td><td>Curved Arrow drawn as shown:</td><td> 
</td></tr>
<tr><td>4a</td><td>94 N</td><td> 

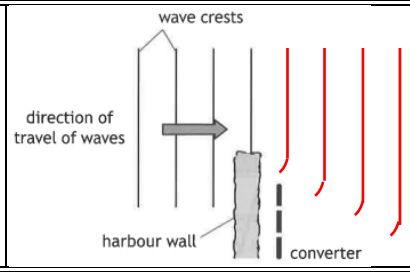
From graph:
When altitude = 140 km
Gravitational field strength = 1.4 N kg^{-1}
W = ? m = 67 kg g = 1.4 N kg^{-1}
W = m x g (1 mark)
W = 67 x 1.4 (1 mark)
W = 94 N (1 mark)</td></tr>
<tr><td>4b</td><td>one answer from:</td><td>exposure to radiation | pressure differential
lower gravitational field strength | extreme temperatures</td></tr>
<tr><td>4c</td><td>answer to include:</td><td>1 mark: Acceleration increases
1 mark: mass decreases (as fuel is used) or gravitational field strength decreases (with altitude)
1 mark: weight decreases or unbalanced force increases</td></tr>
<tr><td>5</td><td>Open ended question:
Candidate Marks gained:
3 marks 5% | 2 marks 14% | 1 mark 37% | 0 marks 44%</td><td>1 mark: Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.
2 marks: Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.
3 marks: Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be ‘excellent’ or ‘complete’ for the candidate to gain full marks.</td></tr>
<tr><td>6a</td><td>one answer from:</td><td>not affected by weather | no (distortion from) atmosphere | no light pollution | can use telescope during the day | Full range of EM waves can be observed</td></tr>
<tr><td>6b(i)</td><td>A planet outside our solar system</td><td>Exoplanets orbit a star outside our solar system and information about the size and orbit od the star can be worked out from the amount of light arriving from the star as the planet crosses the sun</td></tr>
</table>

<table>
<tr><td>6b(ii)</td><td>working showing
3.9×10^{17} m</td><td>d = ? $\quad$ $\nu = 3.0 \times 10^8$ m s^{-1} $\quad$ t = 41 years = 41 x 365.25x24x60x60 s
d = ν x t (1 mark)
d = 3.0×10^8 x 41 x 365.25 x 24 x 60 x 60 (1 mark)
d = 3.9×10^{17} m (1 mark)</td></tr>
<tr><td>6c</td><td>Helium
+
Nitrogen</td><td> 
Star
Hydrogen [Not in star]
Helium
Mercury [Not in star]
Nitrogen</td></tr>
<tr><td>7a</td><td>electrons flow in one direction (only)</td><td>d.c. current has electrons/negative charges flowing in one direction only.
a.c. current has the direction of electron flow continually changing back and forth,</td></tr>
<tr><td>7b</td><td>13500 C</td><td>Q = ? $\quad$ I = 2.5A $\quad$ t = 1.5 hours = 1.5x60x60s
Q = I t (1 mark)
Q = 2.5 x 1.5x60x60 (1 mark)
Q = 13500 C (1 mark)</td></tr>
<tr><td>7c</td><td>Answer to include:</td><td><table><tr><td>1 mark</td><td>(As the temperature increases) the voltage across the thermistor decreases</td></tr><tr><td>1 mark</td><td>Voltage across the resistor increases</td></tr><tr><td>1 mark</td><td>MOSFET turns on</td></tr></table></td></tr>
<tr><td>7d</td><td>160Ω</td><td>Current in series circuit is same at all points = 18mA = 0.018A
Total Voltage across both components in series = 5.0V
Voltage across resistor = supply voltage – voltage across LED
= 5.0V – 2.2 V
= 2.8V
V = 2.8V $\quad$ I = 0.018A $\quad$ R = ?
V = I R (1 mark)
2.8 = 0.018 x R (1 mark)
R = $^{2.8}/_{0.018}$ = 156Ω (1 mark)</td></tr>
<tr><td>8a(i)</td><td>Graph showing:</td><td><table><tr><th>1 mark</th><th>1 mark</th><th>1 mark</th></tr><tr><td>suitable
scales, labels and units</td><td>all points plotted accurately
to ± half a division</td><td>best fit curve</td></tr></table></td></tr>
</table>

<table>
<tr><td>8a(ii)</td><td>6.6 V</td><td> 
</td></tr>
<tr><td>8a(iii)</td><td>Vary the resistance of the variable resistor</td><td>The variable resistor can be used to increase and decrease the resistance in a circuit.
As the resistance is decreased on the variable resistor, the current in the circuit would increase as a result due to Ohm's law. As the resistance of the variable resistor decreases, the voltage across the variable resistor would decrease meaning the voltage across the lamp would increase to maintain the same supply voltage at the battery.</td></tr>
<tr><td>8b</td><td> 
</td><td>A fixed resistor has a constant resistance.
This would give a straight line on the graph as the resistance (gradient) is constant.
Line goes through the origin.</td></tr>
<tr><td>9a(i)</td><td>7800 J</td><td>E_h = ? c = 4180 J kg^{-1} °C^{-1} m = 0.020 kg ΔT= 100°C – 6.3°C = 93.7°C
E = c x m x ΔT (1 mark)
E = 4180 x 0.020 x 93.7 (1 mark)
E = 7833 J (1 mark)</td></tr>
<tr><td>9a(ii)</td><td>32000 J</td><td>E = ? m = 0.014 kg l = 22.6x10^5 J kg^{-1}
E = m x l (1 mark)
E = 0.014 x 22.6x10^5 (1 mark)
E = 31640 J (1 mark)</td></tr>
<tr><td>9a(iii)</td><td>39800 J</td><td>Total Energy = 7800 J + 32000 J = 39800 J</td></tr>
<tr><td>9b(i)</td><td>140000 J</td><td>P = 750W E = ? t = 180s
$P = \frac{E}{t}$ (1 mark)
$750 = \frac{E}{180}$ (1 mark)
E = 750 x 180 = 135000 J (1 mark)</td></tr>
<tr><td>9b(ii)</td><td>One answer from:</td><td>Heat loss to the surroundings. | Not all the energy used by the microwave is transferred to the water.</td></tr>
<tr><td>10a(i)</td><td>639 kPa</td><td>Volume: Constant p_1 = 655 kPa T_1 = 21°C = 294 K
p_2 = ? T_2 = 14°C = 287 K
(1 mark) $\frac{p_1}{T_1} = \frac{p_2}{T_2}$
(1 mark) $\frac{655}{294} = \frac{p_2}{287}$
$p_2 = \frac{655 \times 287}{294}$
(1 mark) p_2 = 639 kPa</td></tr>
</table>

<table>
<tr><td>10a(ii)</td><td>Answer to include:</td><td>
<table>
<tr><td>1 mark</td><td>(As the temperature decreases) the kinetic energy of the particles decreases / particles move slower</td></tr>
<tr><td>1 mark</td><td>Particles collide with the walls of the container less frequently/often or Particles collide with the walls of the container with less force</td></tr>
<tr><td>1 mark</td><td>Pressure decreases</td></tr>
</table>
</td></tr>
<tr><td>10b</td><td>78.1 kg</td><td>
P = 1.02×10^{6} Pa F = ? A = 7.5×10^{-4} m^2
$P = \frac{F}{A}$ (1 mark)
$1.02\times10^{6} = \frac{F}{7.5\times10^{-4}}$
$F = 765$ N
W = 765N m = ? g = 9.8 $N\,kg^{-1}$
$W = m \times g$ (1 mark)
$765 = m \times 9.8$ (1 mark)
$m = {}^{765}/_{9.8} = 78.1$ kg (1 mark)
</td></tr>
<tr><td>11a</td><td>Direction of vibration at right angles to the direction of wave travel/energy transfer</td><td>
<table>
<tr><th>Wave Type</th><th>Description</th></tr>
<tr><td>Transverse</td><td>Direction of vibration at right angles to the direction of wave travel
e.g. all electromagnetic waves and water waves</td></tr>
<tr><td>Longitudinal</td><td>Longitudinal waves have the direction of the vibration in the same direction as the direction of the wave e.g. sound waves</td></tr>
</table>
</td></tr>
<tr><td>11b(i)</td><td>0.14 Hz</td><td>
t = 7.4s f = ?
$t = \frac{1}{f}$ (1 mark) $\therefore\ 7.4 = \frac{1}{f} = 0.14$ Hz (1 mark)
</td></tr>
<tr><td>11b(ii)</td><td>one answer from:</td><td>
<table>
<tr><td>measure (the time for) more waves to pass</td><td>count the number of waves in a longer period of time</td><td>repeat (the measurement) and average</td></tr>
</table>
</td></tr>
<tr><td>11c</td><td>480 kW</td><td>
When amplitude of wave is 1.5m then wave height = 2x 1.5m = 3.0m


</td></tr>
</table>

<table>
<tr><td>11d</td><td>Diagram showing:</td><td>1 mark
diffraction of waves into right shadow region of the harbour wall
1 mark
consistent wavelengths before and after the harbour wall
 
</td></tr>
<tr><td>12a</td><td>The speed of sound (in air)</td><td>To calculate the distance, both the time and the speed/velocity are required.
As lightning is so fast ($3x10^8$ m s^{-1}) that light is assumed to arrive simultaneous to the lightning flash.</td></tr>
<tr><td>12b(i)</td><td>One answer from:</td><td>ray of light is incident at { 90° / right angles / perpendicular } to the lens
ray of light is travelling along the normal</td></tr>
<tr><td>12b(ii)</td><td>equal to</td><td><table>
<tr><th>Property</th><th>Less Dense medium
↓
More Dense Medium
e.g. air → glass</th><th>More Dense medium
↓
Less Dense Medium
e.g. glass → air</th></tr>
<tr><td>Wave Speed</td><td>deceases</td><td>increases</td></tr>
<tr><td>Change in Wavelength</td><td>deceases</td><td>increases</td></tr>
<tr><td>Change of Direction</td><td>bends towards the normal</td><td>bends away the normal</td></tr>
<tr><td>Frequency</td><td>same</td><td>same</td></tr>
<tr><td>Colour of Light</td><td>same</td><td>same</td></tr>
</table>
The frequency of the red light entering the lens is the same as it is inside the lens and also unchanged when it emerges out the other end of the lens.
The wave speed and wavelength would both decrease as the red ray enters the lens but would be back to the same values when it reemerges from the lens.</td></tr>
<tr><td>12c</td><td>Answer to include:</td><td><table>
<tr><td>1 mark</td><td>(The amplitude of the second bang is) less than (the amplitude of the first bang)</td></tr>
<tr><td>1 mark</td><td>(The reflected sound has travelled further so) the second bang has less energy than the first bang</td></tr>
</table></td></tr>
<tr><td>13</td><td>Open ended question:
<table>
<tr><td colspan="4">Candidate Marks gained:</td></tr>
<tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr>
<tr><td>10%</td><td>25%</td><td>45%</td><td>21%</td></tr>
</table></td><td><table>
<tr><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr>
<tr><td>Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.</td><td>Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.</td><td>Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be ‘excellent’ or ‘complete’ for the candidate to gain full marks.</td></tr>
</table></td></tr>
</table>

2024 Nat5 Physics Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	E	Missing	Invalid
1	E	13	7	13	1	**64**	1	0
2	B	4	**75**	6	11	3	1	0
3	E	5	3	8	24	**58**	1	0
4	B	6	**64**	15	5	7	2	0
5	A	**63**	2	2	29	3	1	0
6	D	2	4	4	**39**	50	1	0
7	C	1	5	**91**	1	0	1	0
8	D	2	7	3	**81**	7	1	0
9	A	**47**	17	10	17	7	1	0
10	B	14	**43**	18	15	8	2	0
11	E	2	8	16	11	**63**	1	0
12	A	**28**	29	15	17	10	1	0
13	B	14	**65**	2	7	11	1	0
14	C	3	10	**86**	0	1	1	0
15	B	6	**84**	4	3	2	1	0
16	D	2	4	13	**62**	18	1	0
17	D	4	7	4	**52**	32	1	0
18	A	**77**	3	7	7	4	1	0
19	C	4	17	**74**	2	2	1	0
20	C	20	8	**57**	7	7	1	0
21	A	**64**	18	4	4	10	1	0
22	B	6	**79**	7	5	2	1	0
23	E	14	2	10	4	**69**	1	0
24	C	20	5	**63**	7	4	1	0
25	B	17	**37**	14	28	2	1	0

2024 Nat5 Physics Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.3	27.9%	25	0.2%
Q1aii	1	0.8	80.7%	25	0.3%
Q1b	3	2	66.5%	52	0.6%
Q1c	3	2	68.2%	61	1.3%
Q2aiA	2	1.2	60.7%	58	4.6%
Q2aiB	2	0.9	45.4%	62	8.1%
Q2aii	3	1.5	51.3%	57	5.1%
Q2b	1	0.7	73.3%	28	4.5%
Q2ci	3	1.7	55.1%	41	1.2%
Q2cii	1	0.3	30.9%	48	17.1%
Q3ai	4	2.7	68.0%	46	1.7%
Q3aii	3	1.9	62.2%	52	8.4%
Q3b	1	0.6	63.5%	24	2.1%
Q4a	3	2.5	81.8%	53	1.1%
Q4b	1	0.4	38.3%	28	2.8%
Q4c	3	0.6	19.8%	37	4.5%
Q5	3	0.8	26.7%	47	12.3%
Q6a	1	0.7	68.8%	37	2.0%
Q6bi	1	0.6	61.9%	30	3.3%
Q6bii	3	2.1	68.8%	62	4.9%
Q6c	1	0.9	89.9%	27	0.3%
Q7a	1	0.5	45.5%	45	3.0%
Q7b	3	2.6	85.4%	53	1.8%
Q7c	3	0.7	24.8%	59	8.4%
Q7d	4	2.3	57.7%	67	3.4%
Q8ai	3	1.9	62.4%	42	0.7%
Q8aii	1	0.7	66.3%	34	2.7%
Q8aiii	1	0.3	31.3%	44	7.0%
Q8b	1	0.6	63.4%	31	5.0%
Q9ai	3	1.9	63.7%	67	3.3%
Q9aii	3	2.1	70.1%	63	7.1%
Q9aiii	1	0.3	29.0%	49	14.9%
Q9bi	3	2.6	87.9%	51	4.9%
Q9bii	1	0.4	43.0%	49	13.3%
Q10ai	3	1.5	50.2%	58	5.2%
Q10aii	3	1.6	53.2%	61	8.0%
Q10b	4	2.3	58.0%	64	10.1%
Q11a	1	0.2	21.9%	41	5.9%
Q11bi	3	2.5	84.2%	52	3.5%
Q11bii	1	0.4	39.8%	33	5.8%
Q11c	1	0.3	34.0%	42	3.1%
Q11d	2	0.8	42.2%	41	4.2%
Q12a	1	0.3	27.2%	22	3.1%
Q12bi	1	0.2	17.6%	36	7.7%
Q12bii	1	0.3	32.6%	15	3.2%
Q12c	2	0.6	28.9%	39	12.8%
Q13	3	1.2	41.2%	57	4.9%
Q14a	3	2	66.7%	53	6.0%
Q14bi	1	0.5	54.2%	55	10.6%
Q14bii	3	2.1	69.8%	63	12.2%
Q15a	1	0.2	23.6%	26	7.5%
Q15b	1	0.2	20.8%	31	14.1%
Q15c	2	0.5	26.0%	52	13.0%
Q15d	1	0.3	25.8%	42	9.6%
Paper	110	60.3	54.8%		

2024 Nat5 Physics Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4
Q1ai	72	28			
Q1aii	19	81			
Q1b	4	40	8	48	
Q1c	22	14	3	62	
Q2aiA	20	39	41		
Q2aiB	35	39	26		
Q2aii	10	31	54	5	
Q2b	27	73			
Q2ci	4	56	9	30	
Q2cii	69	31			
Q3ai	5	32	3	8	53
Q3aii	29	11	4	56	
Q3b	37	63			
Q4a	3	17	11	69	
Q4b	62	38			
Q4c	64	19	10	7	
Q5	44	37	14	5	
Q6a	31	69			
Q6bi	38	62			
Q6bii	16	16	15	54	
Q6c	10	90			
Q7a	54	46			
Q7b	5	3	23	69	
Q7c	58	21	10	11	
Q7d	8	37	3	19	32
Q8ai	12	14	49	25	
Q8aii	34	66			
Q8aiii	69	31			
Q8b	37	63			
Q9ai	15	18	27	40	
Q9aii	21	11	5	63	
Q9aiii	71	29			
Q9bi	8	3	8	82	
Q9bii	57	43			
Q10ai	17	45	8	30	
Q10aii	23	22	27	28	
Q10b	18	30	2	2	48
Q11a	78	22			
Q11bi	7	7	13	73	
Q11bii	60	40			
Q11c	66	34			
Q11d	48	20	32		
Q12a	73	27			
Q12bi	82	18			
Q12bii	67	33			
Q12c	58	25	16		
Q13	21	45	25	10	
Q14a	21	12	13	54	
Q14bi	46	54			
Q14bii	19	12	10	59	
Q15a	76	24			
Q15b	79	21			
Q15c	65	19	17		
Q15d	74	26			

2024 Nat5 Physics Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.9	87.2%	27
2	3	1.5	50.9%	44
3a	1	0.5	52.3%	25
3b	1	0.9	90.3%	18
3c	1	0.7	68.6%	31
3d	1	0.6	58.0%	31
3e	1	0.9	91.6%	32
3f	1	0.8	84.9%	30
4a	1	0.9	92.8%	23
4b	1	0.9	90.4%	27
4c	1	0.9	89.4%	30
4d	1	0.3	33.6%	30
5	1	0.6	57.6%	41
6	1	0.5	54.3%	31
7	2	0.7	34.0%	25
8a	1	1	97.1%	11
8b	1	1	98.1%	21
Paper	20	13.7	68.3%	

Item	Max Mark	0	1	2	3
1	1	13	87		
2	3	11	40	33	15
3a	1	48	52		
3b	1	10	90		
3c	1	31	69		
3d	1	42	58		
3e	1	8	92		
3f	1	15	85		
4a	1	7	93		
4b	1	10	90		
4c	1	11	89		
4d	1	66	34		
5	1	42	58		
6	1	46	54		
7	2	60	13	28	
8a	1	3	97		
8b	1	2	98		

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation.
2	3	An account of physics relevant to the aim of the investigation.
3a	1	A brief description of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in a correctly produced table
3d	1	Mean and/or derived values calculated correctly
3e	1	Data relevant to the experiment from an internet/literature source
3f	1	A reference for the source of the internet/literature data
4a	1	An appropriate format from the options of scatter, line or bar graph.
4b	1	The axis/axes of the graph has/have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit
5	1	Valid comparison of the experimental data with data from internet/literature source.
6	1	A conclusion that relates to aim and supported by all the data in the report.
7	2	An evaluation of the experimental procedure.
8a	1	An informative title.
8b	1	A clear and concise report.